Properties of rectangles, rhombuses, and squares

1

a 16 m **b** 90°

2

a 5 m **b** 5 m **c** 5 m

3 90°

4

a 90° **b** 60° **c** 60° **d** 120°

5

a 19 cm **b** 38 cm **c** 90° **d** 45°

6

a 10 **b** 40°

7

a 17 **b** 130°

8

a

i Rhombus **ii** $x = 5$ **iii** $y = 11$

b

i Square **ii** $x = 4$ **iii** $y = 9$

9

a 70 **b** 120

10

a 90° **b** 27° **c** 54° **d** 126°

e 63° **f** 63°

11

a True **b** False **c** False **d** False

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To prove: $ABCD$ is composed of four congruent triangles

	Statements	Reasons
1.	$ABCD$ is a rhombus	Given
2.	$ABCD$ is a parallelogram $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Definition of a rhombus
3.	$\overline{AE} \cong \overline{EC}$ and $\overline{DE} \cong \overline{EB}$	If a quadrilateral is a parallelogram, then its diagonals bisect each other.
4.	$\overline{AE} \cong \overline{AE}, \overline{BE} \cong \overline{BE}$, $\overline{CE} \cong \overline{CE}$, and $\overline{DE} \cong \overline{DE}$	Reflexive property of congruence
5.	$\triangle AEB \cong \triangle CEB \cong \triangle CED \cong \triangle AED$	Side-side-side congruence theorem

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a True **b** False **c** False **d** False

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To prove: $\overline{AC} \cong \overline{BD}$

	Statements	Reasons
1.	$ABCD$ is a rectangle	Given
2.	$ABCD$ is a parallelogram $\angle ADC$ and $\angle BCD$ are right angles	Definition of a rectangle
3.	$\angle ADC \cong \angle BCD$	All right angles are congruent.
4.	$\overline{AD} \cong \overline{BC}$	If a quadrilateral is a parallelogram, then its opposite sides are congruent.
5.	$\overline{DC} \cong \overline{DC}$	Reflexive property of congruence
6.	$\triangle ACD \cong \triangle BDC$	Side-angle-side congruence theorem
7.	$\overline{AC} \cong \overline{BD}$	Corresponding parts of congruent triangles are congruent (CPCTC).

Additional questions

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a False **b** True **c** False **d** False

e True **f** True

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a The quadrilateral is a square. A quadrilateral with one right angle and four congruent sides is a square by definition.

b The quadrilateral is a rhombus. A quadrilateral with four congruent sides is a rhombus by definition.

c The quadrilateral is a parallelogram. A quadrilateral with opposite sides and angles congruent is a parallelogram.

d The quadrilateral is a rhombus. A quadrilateral with opposite sides parallel and consecutive sides congruent is a rhombus.

17 $m\angle 1 = 24^\circ$
$m\angle 2 = 48^\circ$
$m\angle 3 = 66^\circ$
$m\angle 4 = 66^\circ$
$m\angle 5 = 24^\circ$
$m\angle 6 = 66^\circ$

18 $m\angle 1 = 54^\circ$
$m\angle 2 = 36^\circ$
$m\angle 3 = 90^\circ$
$m\angle 4 = 54^\circ$

19

a $x = 7$

b $m\angle BAC = 37^\circ$

c $m\angle ABC = 106^\circ$

20

a Rectangle

b $x = -\frac{40}{3}$

21

a Based on the markings, quadrilateral E" is not guaranteed to be a rectangle. There is not enough information given to conclude the shape is a rectangle.

b If quadrilateral E had congruent diagonals or at least one right angle then it would be a rectangle.

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22 Quadrilateral 1:

a Square

b We can determine if it is a rectangle if the four segments formed by the intersection of the diagonals are the same length. In addition, if all four sides of the rectangle are all equal length then the rectangle would be a square.

Quadrilateral 2:

a Parallelogram

b We can determine if it is a parallelogram if the two pairs of opposite sides are the same length.

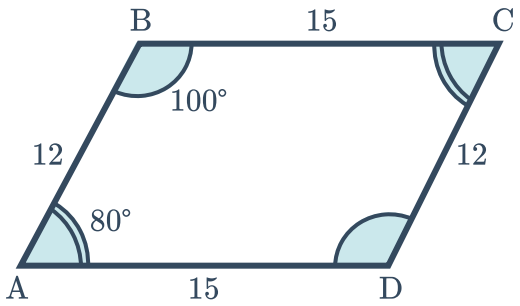
Quadrilateral 3:

a Square

b We can use a similar measurement as in Quadrilateral 1.

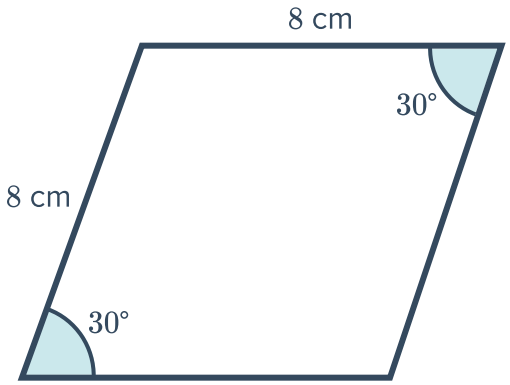
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a Any parallelogram that doesn't have any right angles, and consecutive sides are different lengths. For example:

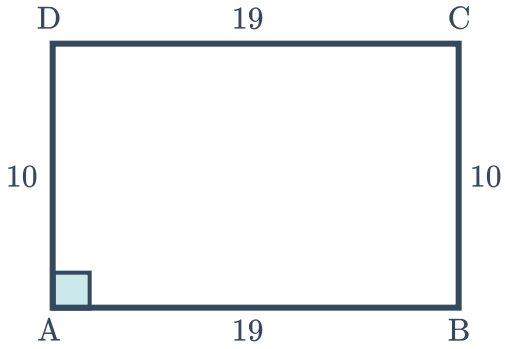


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b Any figure that is a rhombus. For example:



c Any figure that is a rectangle (including squares). For example:



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a There is not enough information to conclude that parallelogram $CDEF$ is a rhombus. In order to be a rhombus we would need to know that one pair of consecutive sides are congruent, the diagonals are perpendicular, or that the diagonal(s) bisect the angles.

b There is enough information to conclude that parallelogram $NOPQ$ is a rectangle. If the diagonals of a parallelogram are congruent, then the parallelogram is a rectangle.

c There is enough information to conclude that parallelogram $EFGH$ is a rhombus. Since $x = 3$, we get that $\angle HGE = 24^\circ$ and $\angle FGE = 24^\circ$. So the diagonal $\overline{EG}$ bisects $\angle HGF$, meaning the parallelogram is a rhombus.

25 Since $\overline{AB} \cong \overline{CD}$, we can conclude that $x = 11$, meaning that $\overline{BC} = 31$. Since $AD \neq BC$, $ABCD$ is not a parallelogram and thus is not a rhombus.

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